

TL;DR. Across three model families (Llama, Gemma and Qwen), we analyze a small set of attention heads that are shared by the circuits of six tasks in each model (Addition, Boolean Logic, IOI, CopyColors MCQA, ARC (Easy) and ARC (Challenge)). Most of these heads have none of the roles known from prior work, or only roles that do not depend on the task. However, in each model we find exactly one shared head that acts as both a name-mover head in IOI and a correct-letter head in the multiple-choice tasks. Surprisingly, a head with this specific combination of roles appears in all three model families. Through a series of correlational and causal analyses, we characterize this head as an *answer mover*: it attends to the answer the model is about to give and moves abstract information about it to the final position. These findings lead us to conclude that most of the overlap between task circuits does not reflect a shared function, but that some shared components do implement a common function, one that recurs across tasks and model families.

1 Case Study: Shared Attention Heads across Task Circuits

In the original paper ("How Much Do Circuits Tell Us?"), we found that the circuits of different tasks overlap heavily at the level of components, in every model we study. This raises a natural question: do the shared components perform the same computation in every task, or do they appear in every circuit for other reasons? We consider two hypotheses. *Hypothesis 1* is that a shared attention head implements a common function across tasks. *Hypothesis 2* is that a shared attention head does not implement a common function, and appears in every task circuit for other (potentially spurious) reasons.

To test these hypotheses, we perform a mechanistic analysis of the attention heads that are shared by all six task circuits. We begin by measuring how many attention heads are shared (Section 1.1) and how much pairs of task circuits overlap (Section 1.2). We then score every attention head on a set of known roles from prior literature, and discover that most shared heads have none of these roles, and that those that do mostly have roles that do not depend on the task, such as previous-token heads. However, we are able to identify a single shared attention head that acts as both a name-mover head in IOI and a correct-letter head in the multiple-choice tasks. Surprisingly, we find a head with this specific combination of roles appears among the shared heads of every model family we study, although its importance for the model’s final answer varies across models (Sections 1.3 and 1.5).

We then study the mechanism of this head in Llama-3.2-3B in detail, and test how well our understanding of its function generalizes to new (constructed) tasks, for which we did not explicitly identify circuits (Section 1.4). Finally, we observe that EAP-IG and RelP circuits yield different shared attention heads, and investigate why this is the case (Section 1.6). We choose not look at the role of MLP blocks, since our results with MLP neurons suggest that a full MLP block is too large a unit for this analysis.

Overall, the evidence is largely consistent with *Hypothesis 2*. Most shared attention heads have no identifiable role, and which heads are shared depends on the attribution method. The exception is a single head, recurring across model families, that attends to the answer and moves information about it to the final position in several tasks (e.g. IOI and CopyColors MCQA).

1.1 How many attention heads are shared?

Table 1 shows the number of attention heads that appear in the top- K circuits of all six tasks at $P = 50\%$. We refer to these as the *shared attention heads*. For most models, the set of shared attention heads is non-empty from $K = 10\%$ onward. We observe that ranking components by

EAP-IG scores generally yields more shared attention heads than ranking them by RelP scores. Additionally, the two methods do not always share the same heads. We therefore analyze the shared attention heads of both methods for the rest of this section.

Model	Attribution Method	K				
		5%	10%	20%	30%	50%
Llama-3.2-3B	EAP-IG	0	3	15	52	227
	RelP	0	3	8	12	53
Llama-3.2-3B-Instruct	EAP-IG	0	1	15	53	218
	RelP	1	3	6	10	54
Gemma-2-2B	EAP-IG	0	0	2	14	44
	RelP	0	1	3	5	15
Gemma-2-2B-IT	EAP-IG	0	0	2	9	46
	RelP	0	2	3	6	20
Qwen3-4B	EAP-IG	1	11	36	93	298
	RelP	3	5	10	16	102

Table 1: **Number of shared attention heads.** For each model and attribution method, the number of attention heads that appear in the top- K circuits of all six tasks at $P = 50\%$.

1.2 How much do pairs of task circuits overlap?

One reason the six-way intersection is small is that it requires a head to appear in the circuits of tasks with very different structure, such as copying a name from the context in IOI and computing a sum in Addition. To measure overlap without this requirement, we compute, for pairs of tasks, the fraction of the attention heads of the smaller task circuit that also appear in the other task circuit, at $K = 10\%$ and $P = 50\%$ (Fig. 1).

Related tasks share most of their attention heads. The circuits of ARC (Easy) and ARC (Challenge) share 87 to 100% of their heads, and those of CopyColors MCQA and ARC (Easy) share 47 to 100%. The IOI circuit shares 35 to 67% of its heads with that of CopyColors MCQA. In contrast, Addition shares only 9 to 45% of its heads with the other tasks, Boolean Logic shares 28 to 60%, and the two share 12 to 34% with each other. These two tasks therefore account for most of the drop from pairwise overlap to the six-way intersection.

1.3 Do the shared attention heads have known roles?

If *Hypothesis 1* holds, the shared attention heads should perform identifiable functions. To test this, we score every attention head of each model on a set of known roles from prior literature. We use the attention weights from the final position on 200 prompts per task (50 for CopyColors MCQA), and on 100 sequences of 25 random tokens repeated twice. We consider the following roles, measuring attention from the final position:¹

¹To decide what counts as attending strongly or mostly to a position, we threshold each statistic based on its distribution across the model’s heads, which identifies the heads whose attention to that position is far above that of a typical head.

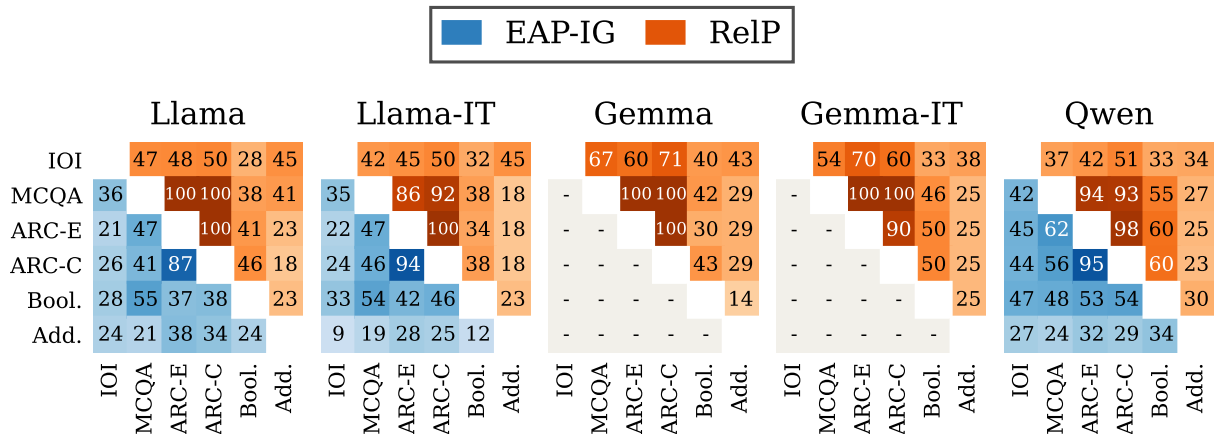


Figure 1: **Pairwise overlap of task circuits.** For each model and pair of tasks, the percentage of the attention heads of the smaller task circuit that also appear in the other task circuit, at $K = 10\%$ and $P = 50\%$, for EAP-IG (below the diagonal) and RelP (above the diagonal). The EAP-IG task circuits of the Gemma models contain at most four attention heads, so we omit them (dashes).

- A *name-mover head* (Wang et al., 2023) attends strongly to the indirect object in IOI, more than to the subject.
- A *correct-letter head* (Lieberum et al., 2023; Wiegrefe et al., 2025) attends strongly to the letter or final newline of the option the model predicts in the multiple-choice tasks.
- An *amplifier* attends strongly to its own position and has a large direct effect on the logit difference in the multiple-choice tasks, similar to the amplification heads of Lieberum et al. (2023).
- *Previous-token*, *duplicate-token* and *induction* heads (Olsson et al., 2022; Wang et al., 2023) mostly attend to the previous token, to the earlier copy of the current token, or to the token after that copy, respectively, in the repeated sequences.
- *Attention sinks* (Xiao et al., 2024) attend almost only to the beginning-of-text token.

Since the models differ in their number of heads, we say that a head attends strongly to a position, or has a large direct effect, when it is in the top 5% of the model’s heads on that statistic.

Most shared attention heads have none of these roles. Fig. 2 and Table 2 show the fraction of shared attention heads with each role. In every model, most shared attention heads have none of these roles, at a rate similar to that of all heads of the model (77 to 84%). This is more pronounced under EAP-IG, where 67 to 100% of the shared attention heads have no identified role. The roles that do surface are mostly previous-token, duplicate-token or induction heads, which are mostly identified by the random repeated sequences rather than the original tasks, and late amplifiers (in the Llama models and Qwen3-4B), which simply increase the logit of the model’s already chosen prediction. In other words, they do not seem to be doing anything task-specific.²

²Note that a head labeled here as having “no role” is not necessarily uninterpretable, since our list of roles is not exhaustive.

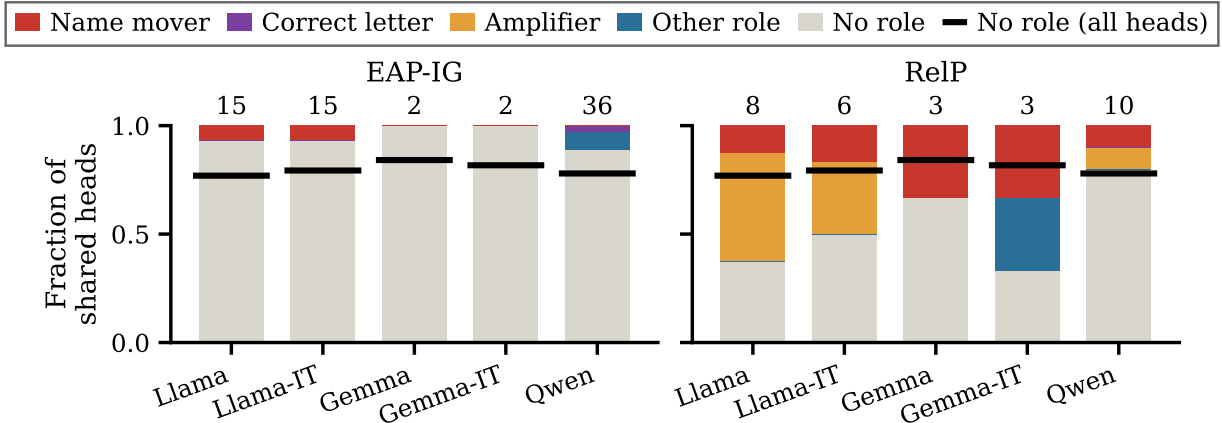


Figure 2: **Roles of the shared attention heads.** For each model and attribution method, the fraction of the shared attention heads ($K = 20\%$, $P = 50\%$) with each role. Black lines mark the fraction of all heads of the model with no role, and the number of shared attention heads is given above each bar.

Model	All heads	EAP-IG			RelP		
		$K = 10\%$	20%	30%	$K = 10\%$	20%	30%
Llama-3.2-3B	0.77	0.67 (3)	0.93 (15)	0.92 (52)	0.33 (3)	0.38 (8)	0.50 (12)
Llama-3.2-3B-Instruct	0.79	1.00 (1)	0.93 (15)	0.96 (53)	0.67 (3)	0.50 (6)	0.60 (10)
Gemma-2-2B	0.84	– (0)	1.00 (2)	0.71 (14)	0.00 (1)	0.67 (3)	0.60 (5)
Gemma-2-2B-IT	0.82	– (0)	1.00 (2)	0.78 (9)	0.50 (2)	0.33 (3)	0.67 (6)
Qwen3-4B	0.78	0.82 (11)	0.89 (36)	0.88 (93)	0.80 (5)	0.80 (10)	0.88 (16)

Table 2: **Shared attention heads without a known role.** The fraction of the shared attention heads ($P = 50\%$) with none of the roles in Section 1.3, compared with the fraction among all heads of the model. The number of shared attention heads is given in parentheses.

The same cross-task head appears in every model family. However, we are able to identify a single shared attention head that acts as both a name-mover head in IOI and a correct-letter head in the multiple-choice tasks. Surprisingly, we find that a shared head with this specific combination of roles appears in every model we study: L15H18 in the Llama models (we write $L\ell Hh$ for attention head h of layer ℓ , with both indexed from 0), L18H6 in the Gemma models and L23H4 in Qwen3-4B (Table 3). We find this somewhat surprising considering these three model families are trained independently (although, speculatively, on the same sort of pre-training distribution). We refer to this head going forward as an *answer mover*, since in both kinds of task it attends to the answer the model is about to output. Our next experiments analyzing this head will further support this characterization. This head doesn’t surface across all methods however: it appears in the shared heads set under both EAP-IG and RelP in the Llama models, but only under RelP in the other models.

1.4 What are the answer-moving heads doing?

The answer movers of Section 1.3 are the strongest candidates for *Hypothesis 1*. We study their mechanism with L15H18 in Llama-3.2-3B, where the answer mover is shared under both attri-

bution methods, and then test our understanding of this mechanism on new synthetic tasks. We compare the effect of these heads on the answer across model families in Section 1.5. L15H18 is in all six EAP-IG task circuits at $K = 10\%$, and it is the only shared attention head whose attention moves to the position of the answer in each task, rather than to a fixed position such as the previous token.

L15H18 attends to the answer the model predicts. Fig. 3 shows the attention of every head of the model on the answer position. In CopyColors MCQA and the two ARC tasks, L15H18 places more attention on the newline that ends the correct option than any other of the model’s 672 heads (0.46 to 0.57 of its attention). In IOI, it ranks third in attention on the indirect object (0.48, compared with 0.03 on the subject). Its attention follows the answer the model gives rather than the correct answer: on the prompts that the model answers incorrectly, it attends most to the predicted option in 75 to 85% of cases and to the correct option in only 10 to 15%.

L15H18 works together with the other two heads of its key-value group. Llama-3.2-3B uses grouped-query attention (?), in which each layer has 24 query heads but only 8 key-value heads, so L15H18 shares its keys and values with L15H19 and L15H20. To measure what each of these heads writes, we compute its direct effect on the logit of a candidate c (its output projected onto the unembedding of c after the final RMSNorm, following Wang et al. (2023)):

$$d_c = W_U[c]^\top \left(\gamma \odot \frac{W_O^h z_T}{\text{rms}(r_T)} \right), \quad (1)$$

where z_T is the head’s output at the final position before the output projection W_O^h , r_T is the residual stream at the final position before the final RMSNorm with gain γ , and $W_U[c]$ is the unembedding of c . We find that L15H18 and L15H19 act as answer movers: they attend to the predicted option and raise its label, with a within-prompt correlation of 0.89 to 0.96 between the attention on an option and the direct effect on its label. In contrast, L15H20 acts as a *distractor suppression* head: it attends to the other candidates and lowers them, clearly in IOI (correlation of -0.84) and weakly in the multiple-choice tasks (-0.21 to -0.41).

The attention of these heads determines the answer. A correlation between attention and direct effect does not establish that the attention causes the answer. To test this, we steer the heads: we set their attention weights from the final position to a one-hot vector on the newline of an incorrect option (or on the first mention of the other name in IOI). Each head then writes the value vector at that position, while the rest of the forward pass is unchanged. Fig. 4 shows the fraction of (prompt, incorrect option) pairs in which the model then predicts the steered-to option. Steering L15H18 alone changes the answer in 48 to 69% of multiple-choice prompts. Steering L15H18 and L15H19 together changes it in 99.5 to 100%, both when comparing the candidates and when taking the model’s greedy completion. For comparison, steering two randomly chosen heads of the same layer has no effect. A weaker intervention, which only moves the heads’ own attention weight from the correct option to the incorrect one, still changes 64 to 73% of answers, and 84 to 94% when L15H20 is moved in the opposite direction. The effect is weaker in IOI: steering all three heads makes the subject beat the indirect object in 92% of prompts, but makes it the greedy completion in only 17.5%.

Removing the heads, however, barely changes the answer. Zeroing L15H18 changes accuracy by at most 2.5 percentage points, and zeroing all three heads by at most 5.

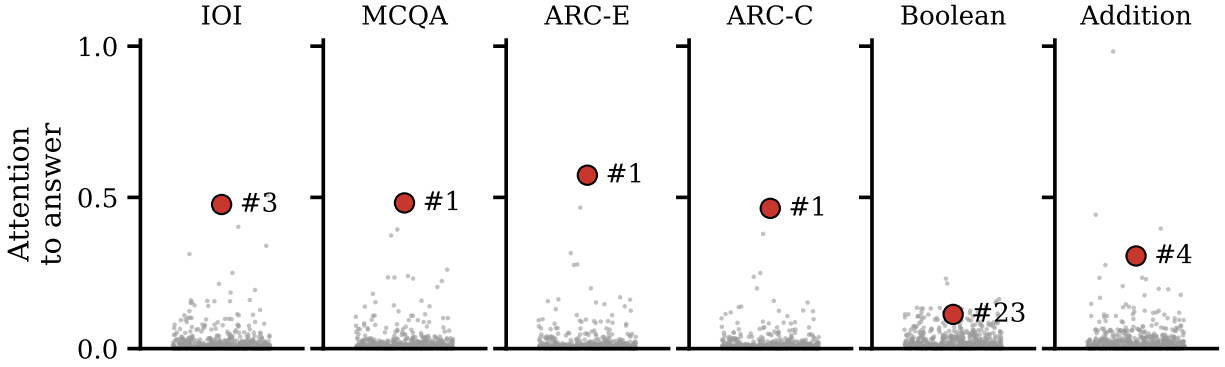


Figure 3: **Attention of every attention head of Llama-3.2-3B on the answer position.** The answer position is the indirect object in IOI, the newline ending the correct option in the multiple-choice tasks, the literals equal to the answer in Boolean Logic, and the first operand in Addition. L15H18 is shown in red and every other head in grey, and the panel titles give the rank of L15H18 among the 672 heads.

The heads generalize to new tasks. To test our understanding of these heads beyond our six tasks, we construct four synthetic tasks, for which we did not identify circuits, in which the answer is a word in the prompt (for example, “The drawer contains the apple. The chest contains the candle. The box contains the stone. The chest contains the”). In these tasks, L15H18 attends more to the answer than to the distractors, and steering the three heads to a distractor changes 84 to 100% of answers.

The heads move information about a word, not its token. We then consider tasks in which the answer must be derived from a word in the prompt rather than copied (“... The bag contains two” → “dogs”). Here, steering the heads to a distractor (“cat”) makes the model output the derived distractor (“cats”) in 98% of prompts and the copied one in 0.2%, compared with 15% for two other heads of the same layer (Fig. 5). This result suggests that the heads move information about the attended word, which later layers transform, rather than copying its token.

The behavior requires candidate answers in context. L15H18 places almost no attention on the subject in subject-verb agreement, steering it does not transfer the operator of an earlier example in arithmetic, and on natural text its attention is less aligned with the positions that most affect the next token than that of most heads. The function of L15H18 is therefore specific to tasks in which the context contains candidate answers, such as the two names in IOI or the lettered options in ARC, rather than general to all tasks.

L15H18 is the only head with both properties. Other heads of Llama-3.2-3B also attend to the answer. The name-mover heads of IOI, such as L18H10 and L18H11, which are not shared attention heads at $K = 10\%$, attend to the letter of the correct option in the multiple-choice tasks (0.44 and 0.55 of their attention), and the sign of their direct effect is the same across tasks. However, their direct effect on the multiple-choice logit difference is small (at most 0.15 in absolute value, compared with 0.39 to 0.58 for L15H18), and steering any one of them changes almost no answers. L15H18 is therefore the only head of the model that both attends to the answer in IOI and has a large effect on the multiple-choice answers.

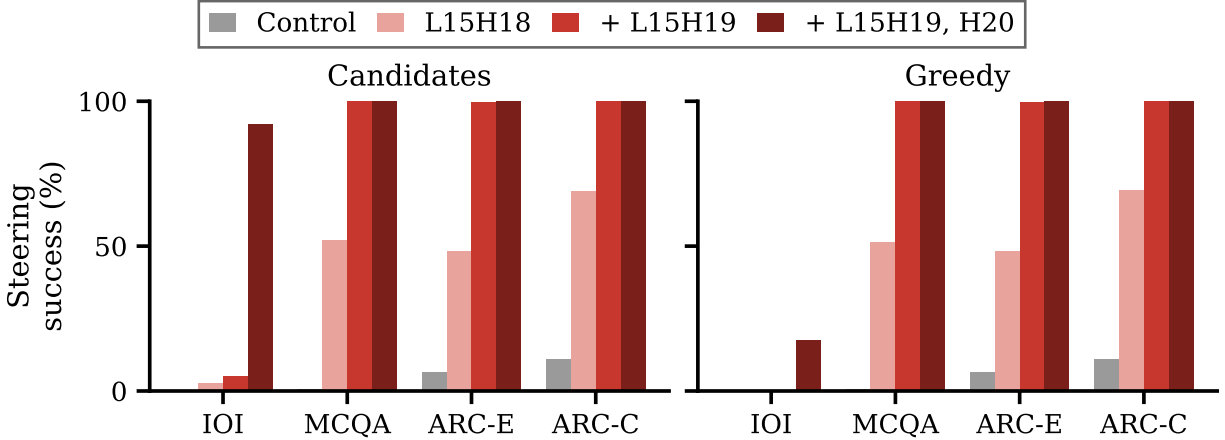


Figure 4: **Steering the key-value group of L15H18.** The fraction of (prompt, incorrect option) pairs in which Llama-3.2-3B predicts the option that the heads are steered to, when steering two control heads of layer 15, L15H18 alone, L15H18 and L15H19, and all three heads of the key-value group (with L15H20 steered to the other candidates). Success is measured as the argmax over the candidates (left) and as the greedy completion over the full vocabulary (right).

1.5 How does the effect of these heads compare across model families?

Table 3 lists, for each model, the head that acts as both a name-mover head in IOI and a correct-letter head in the multiple-choice tasks, together with the effect of steering it. Fig. 6 shows the effect of steering each shared attention head of each model, alone and together with the query heads that share its keys and values, compared with heads drawn at random from the same layers.

Steering the answer mover changes the answer in only some models. Steering the answer-moving head changes many multiple-choice answers in Llama-3.2-3B, Llama-3.2-3B-Instruct and Gemma-2-2B. In Gemma-2-2B, L18H6 is effective when steered to the letter of the option rather than to its final newline. In Gemma-2-2B-IT and Qwen3-4B, however, the answer-moving head changes at most 12% of answers, and no other shared attention head of these models changes more. This suggests that in these models the answer reaches the final position through other components. No single head changes the answer in IOI in any model.

Steering success alone does not identify an answer mover. L10H17 in Llama-3.2-3B places only 0.03 of its attention on the predicted option and has none of our roles. Yet steering it to the newline of an incorrect option changes 60 to 64% more ARC answers than no steering. A likely explanation is that, by layer 10, the value vectors at the end of an option already identify the option, so that any head forced to attend there writes this information. This result serves as a caution. What distinguishes L15H18 is that it attends to the answer without intervention, and that this attention determines the answer when it is changed.

1.6 Why do EAP-IG and RelP share different attention heads?

EAP-IG and RelP circuits yield different shared attention heads. In Llama-3.2-3B, for example, EAP-IG shares L4H10, L9H21 and L15H18 at $K = 10\%$, while RelP shares L23H2, L26H19 and

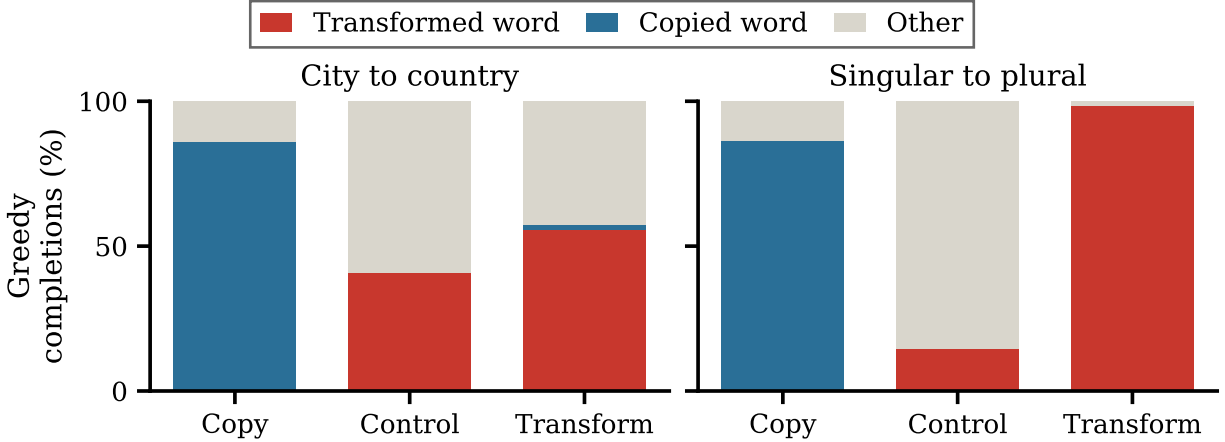


Figure 5: **Copying versus transforming.** The greedy completion of Llama-3.2-3B after steering to a distractor word. *Copy* steers the key-value group of L15H18 in prompts whose answer is a word in the prompt, *Transform* steers it in prompts whose answer is derived from such a word, and *Control* steers two other heads of layer 15 in the latter prompts.

Model	Head	Attention, IOI	Attention, MC	Shared from ($P = 50\%$)	Steering, MC	Steering, IOI
Llama-3.2-3B	L15H18	0.48	0.54	EAP-IG 10%, RelP 20%	0.43–0.58	0.05
Llama-3.2-3B-Instruct	L15H18	0.46	0.53	EAP-IG 20%, RelP 20%	0.62–0.86	0.10
Gemma-2-2B	L18H6	0.65	0.51	RelP 10%	0.25–0.50	0.00
Gemma-2-2B-IT	L18H6	0.64	0.82	RelP 10%	0.00–0.08	0.00
Qwen3-4B	L23H4	0.49	0.14	RelP 20%	0.00–0.12	0.00

Table 3: **Answer-moving attention heads in each model.** Attention heads that act as both a name-mover head in IOI and a correct-letter head in the multiple-choice tasks (MC). Attention is measured on the indirect object in IOI and on the predicted option in the multiple-choice tasks. “Shared from” gives the smallest K at which the head is in all six task circuits. Steering gives the increase in the fraction of (prompt, incorrect candidate) pairs in which the model predicts the steered-to candidate, at the best position and setting, with the range over the three multiple-choice tasks.

L27H1.

Each method favors heads in different layers. One source of this difference is how each method scores a component. EAP-IG (Hanna et al., 2024) attributes edges between components, and we score a component by the sum of the absolute scores of its outgoing edges. This favors early components, since they have many more outgoing edges. RelP (Jafari et al., 2025) attributes each component directly, and our top- K selection keeps its most positive scores, which favors late components that write to the output. Accordingly, the RelP shared attention heads of Llama-3.2-3B are mostly the late amplifiers of Section 1.3: they attend mostly to the beginning-of-text token and to their own position, and raise the logit of the answer the model already predicts, often in several forms of the answer letter. Zeroing L27H1 lowers the multiple-choice logit difference by about 10%.

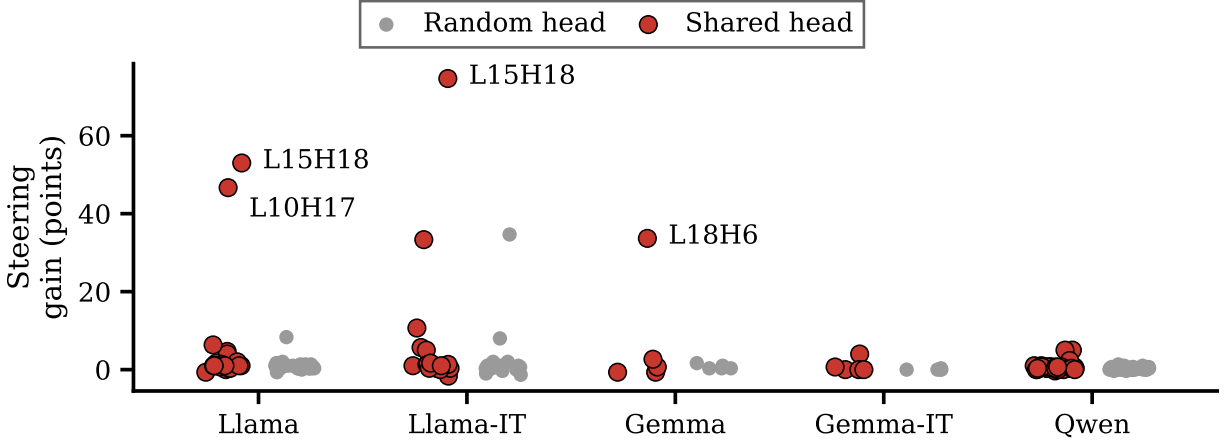


Figure 6: **Steering the shared attention heads.** Each point is a shared attention head (red) or a head drawn at random from the same layer (grey). The vertical axis gives the increase in the fraction of multiple-choice prompts in which the model predicts the steered-to option, at the head’s best steering setting, averaged over the three multiple-choice tasks. In IOI, no head increases this fraction by more than 15 points.

Patching identifies which ranking is closest to the effect of each head. To compare the methods with the effect of each head, we patch the output of each of the 672 heads of Llama-3.2-3B from the counterfactual prompt into the clean prompt, on 100 prompts per task. We then compare the resulting change in the logit difference with three rankings of the heads: EAP-IG with summed edge scores, EAP-IG with scores computed for each head directly, and RelP. As shown in Table 4 and Fig. 7, EAP-IG with direct head scores agrees best with patching: 0.78 of its top 10% of heads are among the top 10% by patching effect, compared with 0.42 for summed edge scores and 0.48 for RelP. Relative to patching, the summed edge scores place too many of their top heads in layers 0 to 6 and too few in layers 21 to 27, while RelP places about twice as many in layers 21 to 27.

Some shared heads reflect how scores are aggregated. L15H18 is among the top 10% of heads in all six tasks under every ranking, and under patching itself. In contrast, L4H10 and L9H21 are shared only under the summed edge scores, which suggests that they reflect this aggregation rather than a role in every task, as *Hypothesis 2* allows. Consistent with this, neither head has a distinctive attention pattern. In particular, neither is an unusual attention sink: the median head of the model places 0.73 of its attention on the beginning-of-text token, compared with 0.85 for L4H10 and 0.42 for L9H21. Moreover, zeroing, mean-ablating or redirecting either head to the beginning-of-text token changes the logit difference by at most 13% in any task.

1.7 Summary

We return to our two hypotheses. The evidence is largely consistent with *Hypothesis 2*. The attention heads shared by all six task circuits are few at strict thresholds, depend on the attribution method and on how its scores are aggregated, and mostly have no identifiable role. Pairs of related tasks share most of their attention heads, but this sharing does not extend to the full set of tasks, and no shared attention head explains the inclusion of Boolean Logic and Addition. The evidence for *Hypothesis 1* is limited to a single head, recurring across model families, that acts as

Ranking	Agreement with patching		Fraction of top heads in layers			
	Top-10% overlap	Spearman ρ	0–6	7–13	14–20	21–27
EAP-IG (summed edge scores)	0.42	0.58	0.29	0.51	0.13	0.07
EAP-IG (head scores)	0.78	0.73	0.18	0.42	0.21	0.19
RelP	0.48	0.58	0.07	0.29	0.30	0.35
Patching	–	–	0.13	0.48	0.21	0.17

Table 4: **Agreement of attribution rankings with patching.** For three rankings of the attention heads of Llama-3.2-3B, the agreement with the effect on the logit difference of patching each head from the counterfactual prompt (averaged over the six tasks), and the fraction of each ranking’s per-example top 10% of heads in each block of layers.

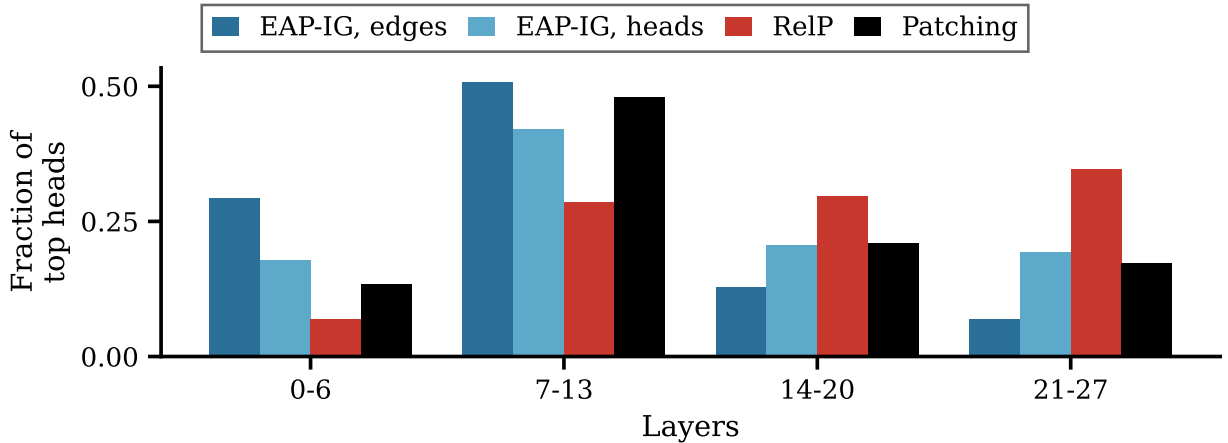


Figure 7: **Layer distribution of the top attention heads.** The fraction of the per-example top 10% of heads of Llama-3.2-3B in each block of layers, under three attribution rankings and under patching each head from the counterfactual prompt.

both a name-mover head in IOI and a correct-letter head in the multiple-choice tasks, attending to the answer and moving information about it to the final position. Steering this head changes the answer only in the Llama models and Gemma-2-2B, and its function appears to be specific to tasks in which the context contains candidate answers.

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